

Distributed Data Analysis and Mining Project Report

Master Degree in Data Science and Business Informatics

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1 Introduction

The dataset used in this study comes from the **Marine Debris Archive (MARIDA)**¹, an open-source dataset designed to detect marine plastic pollution. MARIDA is based on multispectral satellite images acquired by the Sentinel-2 mission. To make these raw images useful for machine learning models, experts visually analyzed and manually annotated the pixels. They classified the data into 15 distinct categories, which include our main target (Marine Debris) along with other marine background features like Sargassum, various water types, clouds, and ships.

The data extraction process was carried out in two main steps. First, we downloaded the core dataset containing the basic spectral signatures of the annotated pixels. This initial data is stored in a tabular format within an HDF5 file (`dataset.h5`).

However, using only the basic spectral bands is not enough to accurately identify marine debris. Therefore, we needed additional features: Spectral Indices (SI) and texture features (GLCM). Since these extra features could not be downloaded directly as a ready-to-use dataset, we had to generate them via code. Using the data engineering scripts provided in the MARIDA repository, we calculated these metrics locally from the original image patches. The newly generated indices and textures were saved into separate files (`dataset_si.h5` and `dataset_glcm.h5`). Finally, these new features were merged with the basic spectral data to build the complete dataset used for our classification and clustering models.

The primary objective of this project is to apply distributed data analysis and mining techniques, specifically leveraging the Apache Spark framework, to large-scale Sentinel-2 satellite imagery for marine debris detection. To systematically address this challenge, our study is guided by a set of interconnected research questions, which are explored through dedicated machine learning tasks. Building on the MARIDA benchmark dataset and its associated baseline results, this work pursues two complementary objectives. First, we explore the dataset’s latent structure through unsupervised learning: by applying distributed clustering algorithms, we aim to segment the data into emergent spectral profiles and, more specifically, to isolate the **Marine Debris** class and identify distinct physical sub-categories, such as dense, sparse, or vegetation-mixed aggregates. Second, we investigate whether the classification performance reported in the

original MARIDA study can be improved by combining supervised learning with imbalanced learning techniques and a fully distributed PySpark pipeline, training models on the complete dataset.

Beyond mere detection, a critical aspect of our analysis is evaluating the model’s spatial generalization capabilities. Through a targeted Domain Shift Analysis, we quantify the impact of geographic variations by training the classifier on a primary region and evaluating its performance on a geographically distinct, unseen domain.

Finally, we utilize Explainable AI (XAI) techniques to determine which specific features are most critical in distinguishing marine debris from complex aquatic confounders.

2 Exploratory Data Analysis

2.1 Dataset Description

The dataset employed in this work consists of multispectral imagery derived from the Sentinel-2 satellite sensor, specifically curated for the classification and monitoring of marine environments. A preliminary quality check confirmed the complete absence of missing values across all feature columns. The input variables are organized into four distinct feature groups:

- **Spectral Signature:** The core optical information of the dataset, comprising the raw reflectance values from 11 spectral bands of the Sentinel-2 sensor.
- **Spectral Indices:** Derived indices computed to highlight specific physical properties. Standard indices such as NDVI (vegetation health) and NDWI (water content) are complemented by marine-specific ones, namely the Floating Debris Index (FDI) and the Floating Algae Index (FAI), designed to distinguish anthropogenic debris from organic matter. Additional indices, NDMI, BSI, and NRD, further separate debris from algae and water backgrounds.
- **Texture Features (GLCM):** Spatial context is captured through features derived from the Gray-Level Co-occurrence Matrix (GLCM), including Contrast, Dissimilarity, Homogeneity, Energy, Correlation, and Angular Second Moment (ASM). These metrics quantify texture patterns, enabling the distinction between the roughness of floating objects and the uniformity of the water surface.
- **Metadata and Target:** Contextual metadata includes the Sentinel-2 Tile identifier and the annota-

¹<https://github.com/marine-debris/marine-debris.github.io>

tor’s confidence level (High, Moderate or Low). The target variable is a categorical label representing one of 15 thematic classes (e.g. **Marine Debris**, **Ships**, **Dense Sargassum**).

2.2 Target Feature Distribution

Beyond analyzing the statistical properties of individual features, it is crucial to examine the distribution of the target variable to understand the structural composition of the dataset.

Class	Count	Percentage (%)
Sediment-Laden Water	372,937	44.54
Turbid Water	157,612	18.82
Marine Water	129,173	15.43
Clouds	117,400	14.02
Shallow Water	17,369	2.07
Cloud Shadows	11,728	1.40
Wakes	8,490	1.01
Waves	5,827	0.70
Ship	5,803	0.69
Marine Debris	3,405	0.41
Dense Sargassum	2,797	0.33
Sparse Sargassum	2,357	0.28
Foam	1,225	0.15
Natural Organic Material	866	0.10
Mixed Water	410	0.05

Table 1: Distribution of the 15 thematic classes in the dataset, expressed in absolute pixel count and relative frequency. Background categories dominate the distribution, while classes of primary interest such as **Marine Debris** and **Sargassum** represent less than 2% of total samples.

As shown in Table 1, the dataset exhibits severe class imbalance. Background categories such as various water types and clouds dominate the distribution, cumulatively accounting for over 92% of the samples. Conversely, the classes of primary interest are extremely rare: **Marine Debris** represents only 0.41% of pixels, followed by **Ships** (0.69%), **Dense Sargassum** (0.33%), and **Sparse Sargassum** (0.28%). This distributional skew poses a significant challenge for standard classification algorithms, making **Marine Debris** one of the most analytically critical and inherently difficult class in the dataset.

2.3 Class Co-occurrence

To further understand the spatial relationships between classes, we analyzed the conditional co-occurrence probability. This analysis reveals the likelihood of Class Y being present in the same image, given the presence of Class X. Heatmap in Figure 1 highlights several strong physical dependencies that validate the dataset’s consistency:

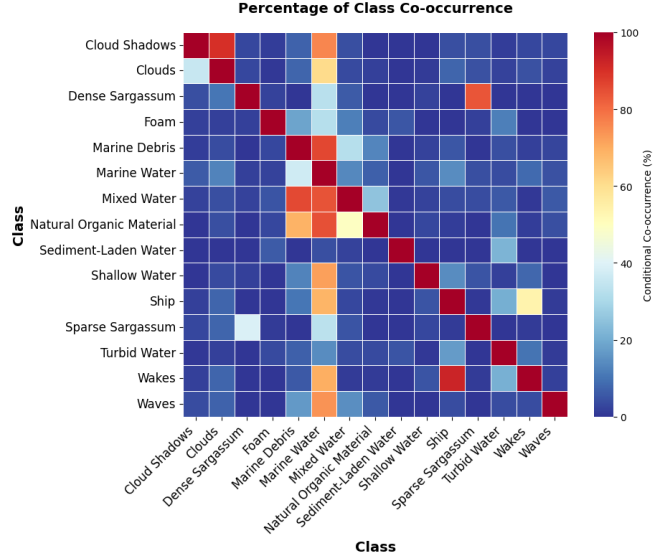


Figure 1: Conditional class co-occurrence matrix computed over the full dataset. Entry (X, Y) denotes the percentage of images containing class X that also contain class Y . Diagonal entries represent self-co-occurrence and equal 100% by definition.

- **Physical Causality:** As expected, transient features show strong correlations with their generating objects. Notably, **Wakes** co-occur with **Ships** in 92.45% of cases, and **Cloud Shadows** co-occur with **Clouds** 90.14% of the time.
- **Target Context (Marine Debris):** The **Marine Debris** class is predominantly surrounded by **Marine Water** (86.33%), confirming its location in open water environments.
- **Critical Associations:** A significant correlation exists between **Mixed Water** and **Marine Debris**. When **Mixed Water** is detected, there is an 85.71% probability that **Marine Debris** is also present. This suggests that the spectral signature of mixed water often characterizes the transitional pixels at the boundaries of floating debris patches.

These results demonstrate that the dataset exhibits a well-defined spatial structure, where class co-occurrences reflect genuine physical and environmental relationships. Such patterns offer a deeper understanding of the inherent complexity of marine scenes, particularly the spectral ambiguity introduced by transitional and boundary pixels.

2.4 City and Confidence Distribution

The dataset comprises imagery from fourteen distinct locations worldwide, ranging from river mouths to open coastal waters. Specifically, the data is sourced from Mo-

tagua, Ulua, La Ceiba, and Roatan (Honduras/Guatemala); La Gonave (Haiti); Santo Domingo (Dominican Republic); Isle of May (United Kingdom); Durban (South Africa); Danang (Vietnam); Jakarta and Bali (Indonesia); Nakdong (South Korea); Yangtze (China); and Manila (Philippines).

However, the spatial distribution exhibits a strong geographical bias. The data is heavily skewed towards Central America: the *Motagua* river mouth alone accounts for 63.6% of the total pixels. When combined with *Ulua* (17.1%) and *La Ceiba* (8.0%), this region represents over 88% of the entire dataset. Conversely, locations in Southeast Asia and Africa, such as *Danang*, *Jakarta*, and *Durban*, are significantly underrepresented, each contributing less than 2% to the training data.

Regarding the **confidence** variable, it indicates the degree of certainty with which an annotator classified a given pixel. Overall, 99.6% of the dataset’s pixels were assigned a high-confidence label, meaning almost all data points were classified with great certainty. However, a deeper analysis of the **Marine Debris** target class reveals a different pattern: only 47.7% of these pixels have a high confidence score, while 36.4% are classified with moderate confidence, and 15.9% with low confidence. This discrepancy highlights that **Marine Debris** pixels are inherently challenging to identify, even for human experts during the manual labeling process.

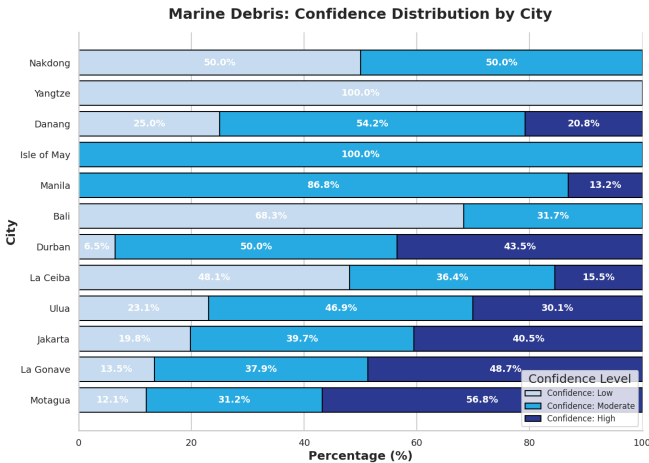


Figure 2: Stacked bar chart showing the distribution of annotator confidence levels for **Marine Debris** pixels, disaggregated by acquisition site. The stark contrast between locations such as *Motagua* (56.8% high confidence) and *Yangtze* or *Bali* (predominantly low confidence) reflects the influence of local environmental conditions on labeling certainty.

Furthermore, this annotation uncertainty is highly region-dependent (Figure 2). For instance, while locations like the *Motagua* river contain a majority of high-confidence marine debris labels (56.8%), other areas such as the

Yangtze river or *Bali* are heavily dominated by low-confidence annotations (100.0% and 68.3%, respectively). This stark regional variability suggests that local environmental conditions, such as water turbidity or debris composition, significantly impact the visibility and unambiguous identification of floating debris by human annotators.

2.5 Emerging Physical Profiles

To evaluate the impact of the spectral and texture indices, we selected the most informative features from each group and aggregated their mean values by class, normalizing them using Z-scores.

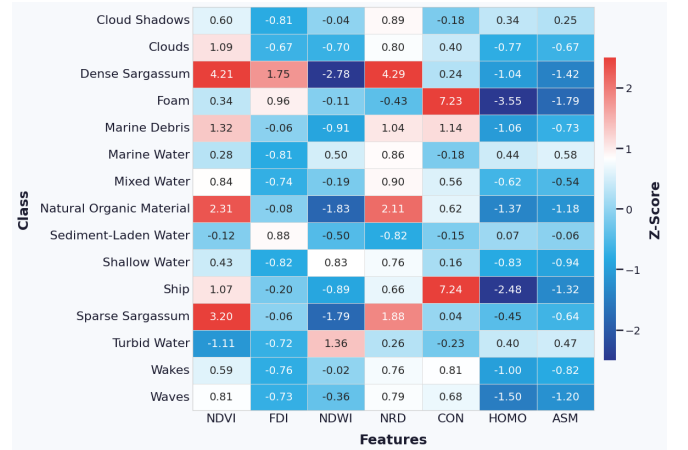


Figure 3: Per-class Z-score profiles of selected spectral indices (NDVI, FDI, NDWI, NRD) and GLCM texture features (CON, HOMO, ASM), computed from mean values aggregated by land cover class. Red cells indicate values significantly above the mean; blue cells indicate values significantly below.

The resulting heatmap (Figure 3) reveals distinct physical profiles that justify our feature selection. Spatial metrics (GLCM) successfully isolate distinct objects from the background: classes characterized by highly irregular surfaces, such as **Ships** and **Foam**, exhibit exceptionally high Contrast (CON = 7.24 and 7.23, respectively) and strongly negative Homogeneity (HOMO = -2.48 and -3.55), representing the most extreme deviations in the entire heatmap. Conversely, uniform surfaces such as **Marine Water** and **Turbid Water** maintain near-average or mildly positive HOMO and ASM values, reflecting their spatially homogeneous texture. Spectral indices effectively highlight specific materials: **Dense Sargassum** dominates with massive positive deviations in both NDVI (+4.21) and NRD (+4.29), while **Sparse Sargassum** follows the same pattern with NDVI = +3.20 and NRD = +1.88, and **Natural Organic Material** also shows notable positive deviations in NDVI (+2.31) and NRD (+2.11). Crucially, these indices work synergistically to disentangle overlapping phenomena: while both **Sediment-Laden Water** and **Sargassum**

present a moderately elevated FDI (indicating floating material at the surface), sediment is effectively separated by its near-zero to slightly negative NDVI (-0.12), in stark contrast to the strongly positive vegetation signal of **Dense** and **Sparse Sargassum**. In particular, the **Marine Debris** class does not exhibit extreme values in any single feature, presenting only mild above-average deviations across NDVI ($+1.32$), NRD ($+1.04$), and CON ($+1.14$). This spectral and textural mixing confirms the inherent difficulty in detecting marine debris, as its signature is consistently overshadowed by the more dominant responses of surrounding classes.

2.6 Features Correlation Analysis

The correlation analysis reveals several key insights into the linear relationships between the engineered features and the raw spectral data.

As shown in Figure 4, the matrix reveals distinct blocks of strongly correlated variables. Most prominently, the raw Sentinel-2 spectral bands, ranging from **nm440** to **nm2200**, form a dense block of near-perfect positive correlations, frequently exceeding 0.90. This multicollinearity indicates that the general reflectance of the observed targets tends to increase or decrease simultaneously across the electromagnetic spectrum. The engineered spectral indices also exhibit coherent inter-relationships: the Shadow Index (SI) reaches an almost perfect negative correlation with the visible spectral bands (-1.00), while NDVI, FAI, and NRD form a tightly correlated cluster (0.70 – 0.94), reflecting their shared sensitivity to vegetation and organic floating matter.

A crucial insight from the matrix is the near-zero correlation between the GLCM texture features and the raw spectral bands. This orthogonality confirms that spatial context, such as surface roughness or pattern uniformity, provides an entirely independent layer of information that spectral signatures alone cannot capture. Within the texture group itself, two distinct and opposing clusters emerge: features describing uniformity, such as Homogeneity (HOMO), Energy (ENER), and Angular Second Moment (ASM), that are highly positively correlated with one another, while they are strongly negatively correlated (up to -0.96) with metrics representing structural complexity, namely Contrast (CON) and Dissimilarity (DIS). This confirms that an increase in image contrast is intrinsically accompanied by a corresponding drop in homogeneity.

3 Data Preprocessing

Following the Exploratory Data Analysis, the initial preprocessing step involved discarding non-informative meta-data. The removed variables fall into two conceptually distinct categories, each with a specific justification:

- **Acquisition identifiers:** **Date**, **Tile**, and **Image** were excluded to prevent data leakage. Since the dataset is heavily concentrated in specific geographic areas and time windows (e.g. the *Motagua* tile alone accounts for 63.6% of all pixels), retaining these variables would allow both clustering and classification algorithms to exploit spatial and temporal regularities in the training data rather than learning the intrinsic physical signatures of each class.
- **Spatial coordinates:** **XCoords** and **YCoords** unlike the identifiers above, do not introduce leakage directly, but their inclusion would distort distance-based computations. In clustering, Euclidean distances in the joint feature space would be dominated by geographic proximity rather than spectral or textural similarity, biasing the resulting groups towards spatial contiguity rather than physical homogeneity.

3.1 Dimensionality Reduction

Building upon the findings from the correlation analysis, we addressed the multicollinearity among the raw spectral bands by applying Principal Component Analysis (PCA), implemented via the distributed **MLlib** library within the PySpark pipeline. The analysis of the cumulative explained variance revealed that the first principal component (PC1) alone captures 92.78% of the total variance, with PC2 bringing the cumulative total to only 97.88%. Given this rapid saturation, the entire spectral signature group was replaced by PC1, significantly reducing dimensionality while preserving the vast majority of the spectral information.

Additionally, four engineered features, namely **SI**, **HOMO**, **ASM**, and **FAI**, were dropped to eliminate redundancy. The Shadow Index (SI) was excluded due to its near-perfect negative correlation with the raw spectral bands, which are already represented by PC1. **HOMO** and **ASM** were removed for their near-perfect positive correlation with retained texture features, while **FAI** was excluded given its strong overlap with NDVI and NRD (respectively 0.70 and 0.94).

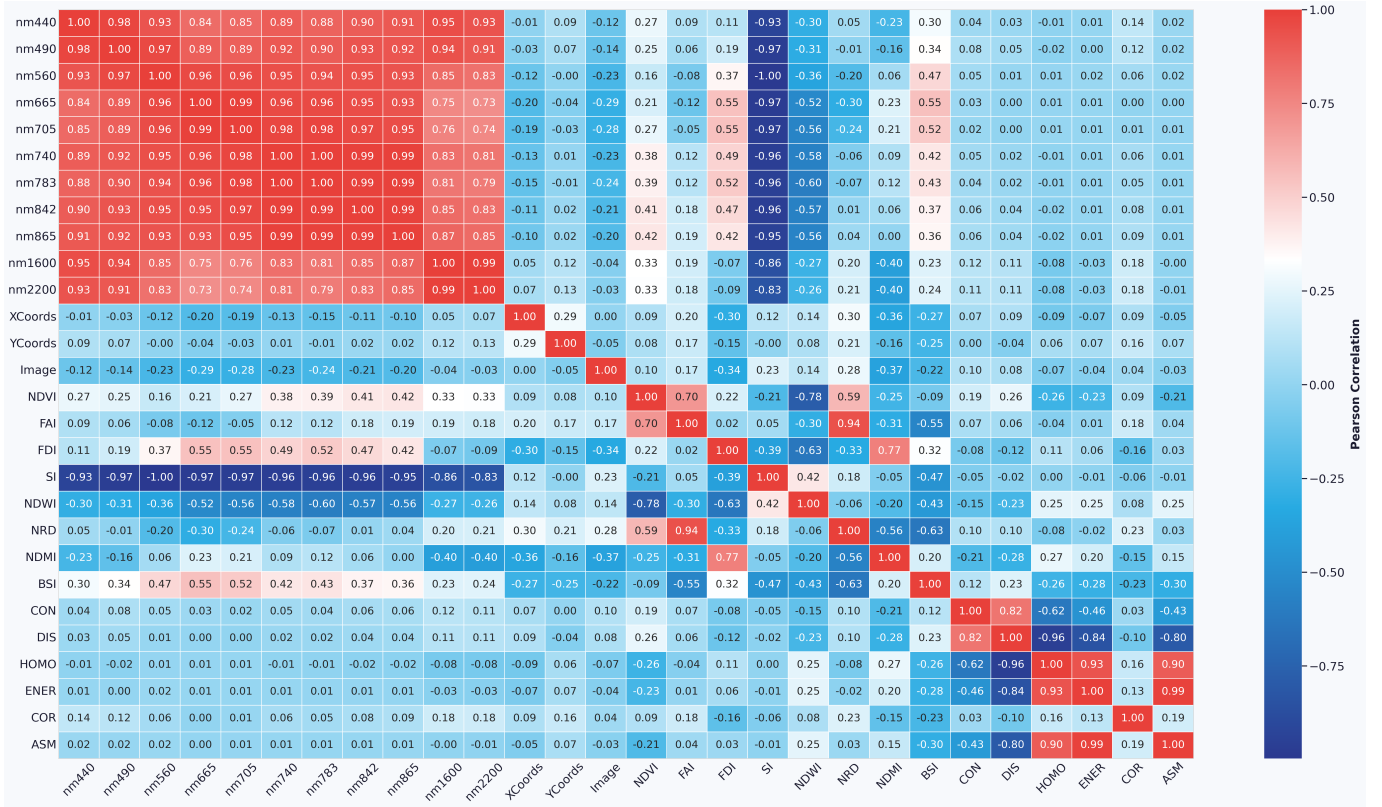


Figure 4: Pearson correlation matrix computed over all numerical variables, encompassing the 11 raw Sentinel-2 spectral bands (nm440–nm2200), 8 engineered spectral indices (NDVI, FAI, FDI, SI, NDWI, NRD, NDMI, BSI), 6 GLCM texture features, and spatial coordinates. Red cells indicate strong positive correlations; blue cells indicate strong negative correlations.

3.2 Outlier Detection and Removal

In the final preprocessing step, outlier detection was performed using a robust statistical approach based on Z-score. For each observation, a Z-score was computed globally across the dataset for each of the 11 numerical features; the outlier score assigned to a row was defined as the maximum absolute Z-score across all features. Any observation whose score exceeded the threshold of 3.5 was flagged as anomalous and removed. This filtering resulted in the exclusion of 41,444 rows (approximately 4.95% of the dataset), yielding a final clean dataset of 795,955 observations.

4 Learning

4.1 Clustering

In this section, we present the unsupervised analysis conducted on the dataset using the distributed K-Means algorithm, natively implemented in PySpark via MLlib. The primary objective of the clustering task is to explore the latent structure of the spectral and texture features. The analysis is structured into three progressive phases: a baseline evaluation setting the number of clusters equal

to the 15 original classes ($k=15$), a search for the optimal global configuration guided by internal validation metrics such as the elbow method and the silhouette score, and finally a targeted sub-clustering exclusively on the **Marine Debris** pixels, in order to extract their latent physical macro-states (e.g., dense, sparse, or mixed aggregates) by neutralizing the interference of the surrounding aquatic environment.

The application of the K-Means algorithm set to $k=15$ aimed to verify whether the latent structure of the data could naturally replicate the 15-class taxonomy defined by the annotators. The results of the contingency table demonstrate that the spectral separability does not correspond to the semantic one. Selected results from the analysis are reported in Table 2 and the values of the centroids are visualized in Figure 5. The analysis of the centroids highlights an extreme fragmentation of the feature space. Observing the values, many clusters show almost overlapping spectral profiles, differing solely by small variations in background-related indices, such as slight fluctuations in NDWI for water or BSI for sediments. This redundancy quantitatively confirms that the algorithm is over-segmenting the surrounding marine environment, wasting numerous groupings to describe

minimal nuances of water brightness or turbidity rather than isolating true, distinct semantic classes like plastic.

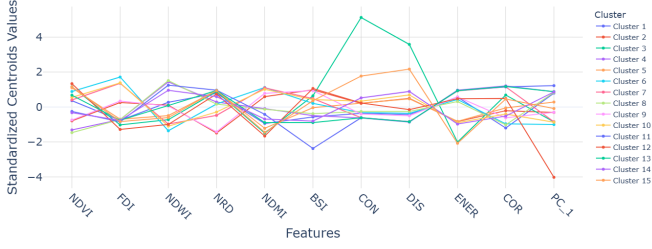


Figure 5: Parallel coordinate plot of the baseline K-Means with $k=15$.

Due to the strong imbalance of the dataset, a monopoly of the background classes is primarily observed: K-Means uses most of the clusters to fragment minimal spectral variations of the dominant water masses. The **Sediment-Laden Water** class alone monopolizes 5 clusters (1, 5, 6, 8, 9) with purities close to 100%, and similar dynamics are observed for **Turbid Water** (clusters 3 and 7) and **Clouds** (cluster 11), leading the algorithm to “waste” over half of the groupings to model the background.

In parallel, there is a strong dispersion of the target class. **Marine Debris** is never effectively isolated, and its pixels are mainly dispersed among clusters 4 (50.74%), 14 (19.46%), and 0 (17.78%). There is no grouping in which **Marine Debris** is the majority component; even in the cluster where it is most concentrated (Cluster 4), its purity barely reaches 2.34%. The analysis of Cluster 4 specifically reveals significant spectral confusion regarding surface anomalies. This cluster acts as an aggregator of floating elements and visual disturbances: it collects a large part of the plastic, but also almost all of the **Dense Sargassum** (84.09%) and **Sparse Sargassum** (70.71%), along with a high percentage of **Waves** (47.63%) and **Wakes** (37.45%), despite being numerically dominated by **Clouds** (56.28%). In summary, the global structure of the data does not reflect the 15 original labels, as the spectral signatures of **Marine Debris** are confused with typical marine confounders.

The reduction of the number of clusters to $k=6$, guided by evaluation metrics visualized in Figure 6, generated a reorganization of the spectral space into much more interpretable latent macro-environments compared to the baseline. Results are summarized in Table 3 and the parallel coordinate plot for the centroids is shown in Figure 7.

The centroids reveal a sharp polarization between aquatic macro-environments (characterized by $NDWI > 1$) and sur-

Cluster	Class	Cluster Purity (%)	Class Dist. (%)
Cluster 0 (6.48%)	Marine Water	49.94	20.80
	Turbid Water	26.91	8.81
	Marine Debris	1.03	17.78
	Other	22.12	-
Cluster 1 (12.43%)	Sediment-Laden Water	99.91	26.51
	Marine Debris	0.00	0.00
	Other	0.09	-
Cluster 2 (2.03%)	Marine Debris	1.64	8.87
	Other	98.36	-
Cluster 4 (8.14%)	Clouds	56.28	39.00
	Marine Water	17.47	9.15
	Marine Debris	2.34	50.74
	Sparse Sargassum	1.44	70.71
	Dense Sargassum	0.23	84.09
	Other	22.24	-
Cluster 14 (3.94%)	Clouds	71.50	24.00
	Shallow Water	7.51	14.53
	Marine Debris	1.85	19.46
	Other	19.14	-
Other 10 Clusters (66.98%)	Marine Debris	0.02	3.15
	Other background	99.98	-

Table 2: Composition of key clusters (Baseline $k=15$). For each cluster, its size relative to the entire dataset is shown in parentheses. The table highlights the dominant classes, the dispersion of the target class (**Marine Debris**), and specific confounders discussed in the text. To provide a complete overview without redundancy, the remaining 10 background-dominated clusters are aggregated. The “Cluster Purity (%)” column indicates the proportion of the cluster composed of a specific class. The “Class Dist. (%)” column shows the percentage of the total pixels of that class assigned to the cluster.

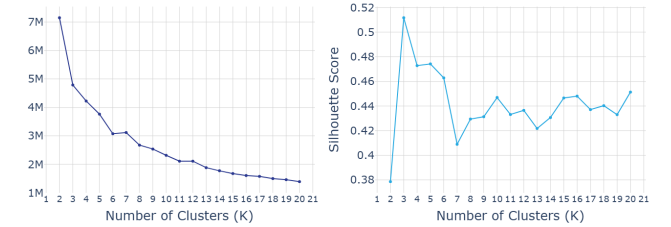


Figure 6: SSE and silhouette scores across different values of k to determine the optimal number of clusters.

face anomalies. Cluster 4 stands out for extreme spatial heterogeneity, reaching maximum values of Contrast (3.31) and Dissimilarity (2.83), typical signatures of chaotic elements like clouds and waves. However, the most significant analytical finding concerns Cluster 1: despite recording the absolute maximum peak of the FDI index (1.57), it consists almost at 99% of **Sediment-Laden Water**. This demonstrates a severe vulnerability in the basic spectral features: the high near-infrared (NIR) reflectance of suspended sediments “fools” the floating debris index, generating false positives mathematically higher than actual plastic.

A rational consolidation of the aquatic background is observed, in which water masses are no longer fragmented into redundant groupings, but are neatly summarized in 4

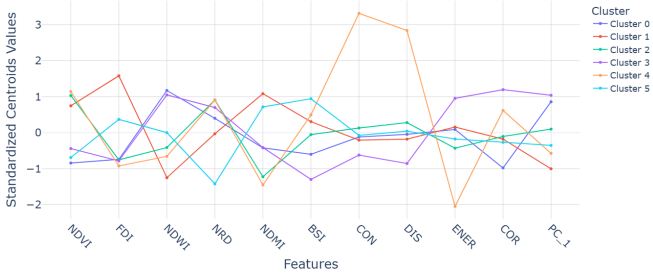


Figure 7: Parallel coordinate plot of the optimized K-Means with $k=6$.

macro-groups. Clusters 1 and 5 almost exclusively capture **Sediment-Laden Water** (purities of 98.9% and 99.95%), Cluster 0 models turbid water by aggregating **Turbid Water** (76.97%) and **Marine Water** (17.8%), while Cluster 3 represents a transition zone composed equally of **Marine Water** (48.12%) and **Turbid Water** (47.87%).

Cluster	Class	Cluster Purity (%)	Class Dist. (%)
Cluster 0 (13.51%)	Turbid Water	76.97	52.57
	Marine Water	17.80	15.47
	Marine Debris	0.03	1.11
	Other	5.20	-
Cluster 1 (20.35%)	Sediment-Laden Water	98.90	42.97
	Marine Debris	0.00	0.13
	Other	1.10	-
Cluster 2 (16.01%)	Clouds	47.30	64.47
	Marine Water	24.73	25.47
	Marine Debris	1.76	75.32
	Sparse Sargassum	0.99	95.90
	Dense Sargassum	0.12	88.07
	Other	25.10	-
Cluster 3 (18.78%)	Marine Water	48.12	58.14
	Turbid Water	47.87	45.46
	Marine Debris	0.02	1.00
	Other	3.99	-
Cluster 4 (5.25%)	Clouds	74.36	33.21
	Waves	4.59	34.83
	Ship	2.60	71.89
	Marine Debris	1.60	22.40
	Other	16.85	-
Cluster 5 (26.10%)	Sediment-Laden Water	99.95	55.68
	Marine Debris	0.00	0.03
	Other	0.05	-

Table 3: Composition of the optimized clusters ($k = 6$). The “Cluster Purity (%)” column indicates the proportion of the cluster composed of a specific class. The “Class Dist. (%)” column shows the percentage of the total pixels of that class assigned to the cluster. The minor classes in each cluster are aggregated into “Other”.

Elements foreign to water masses are confined into two macro-families with distinct physical characteristics. Cluster 2 acts as an aggregator of floating matter, capturing the vast majority of marine vegetation (95.9% of Sparse Sargassum and 88.07% of Dense Sargassum).

Conversely, Cluster 4 collects high-reflectance disturbances and anthropogenic elements, being dominated by **Clouds** (74.36%) and grouping most of the pixels related to **Ships** (71.89%) and **Waves** (34.83%). Regarding the target, although **Marine Debris** cannot form a pure cluster (its maximum purity barely reaches 1.76% in Cluster 2), its dispersion is drastically reduced: 97.7% of the pixels are concentrated in just two groupings, being absorbed for 75.32% by Cluster 2 and for 22.4% by Cluster 4. This optimized configuration demonstrates that the spectral signature of the plastic inevitably overlaps with that of floating vegetation and reflective surface “confounders”, certifying the impossibility of isolating it by analyzing the global dataset and validating the transition to sub-clustering.

The isolation of the **Marine Debris** class and the re-training of a dedicated scaler allowed the K-Means algorithm to focus exclusively on the internal variance of the plastic, neutralizing the background noise of the ocean. Figure 8 reports the Elbow method and Silhouette scores used for optimal k selection. The metrics Distribution of annotator confidence within these sub-clusters is reported in Table 4.

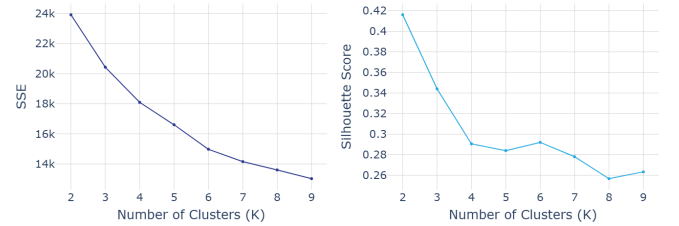


Figure 8: SSE and silhouette scores across different values of k to determine the optimal number of clusters for **Marine Debris** pixels.

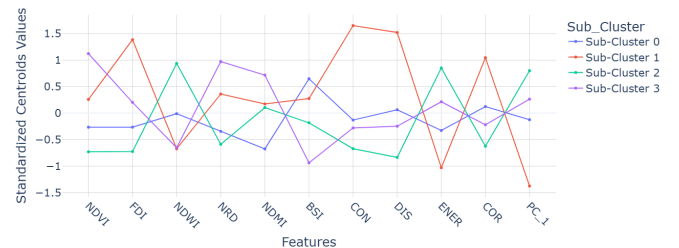


Figure 9: Parallel coordinate plot of the K-Means on Marine Debris pixels only, $k=4$.

The algorithm identified four physical macro-states of the material, whose spectral and texture signatures find direct validation in the **Confidence** distribution of human annotators:

- **Sub-Cluster 1 (Dense and Pure Debris):** Rep-

resents dense plastic on the surface. Spectrally it is characterized by the absolute maximum value of FDI (1.39) and the minimum value of NDWI (-0.67), indicating a total absence of the water signal. The GLCM texture highlights a rough and heterogeneous surface, with maximum peaks of Contrast (1.65) and Dissimilarity (1.52) and the minimum Energy. Being physically unequivocal, it records the highest "High Confidence" from the annotators (64.89%), with minimal uncertainty (Low Confidence at 7.06%).

- **Sub-Cluster 2 (Sparse or Submerged Debris):** It is the exact opposite of the previous one. Spectrally it is dominated by the water signal (maximum peak of NDWI at 0.94) with strongly negative values for NDVI(-0.73) and FDI (-0.72). The texture appears flat and homogeneous (maximum Energy at 0.85, strongly negative Contrast and Dissimilarity). This state generates the maximum human uncertainty: *Low Confidence* reaches its maximum (29.41%), almost equaling the *High Confidence* (30.24%).
- **Sub-Cluster 3 (Debris mixed with Vegetation):** Identifies plastic trapped in floating biomasses. It is unequivocally signaled by an extreme peak in the NDVI vegetation index (1.12) and high values of NRD (0.97) and NDMI (0.72). The structural presence of biomass makes the identification sufficiently reliable, dividing between *Moderate Confidence* (43.97%) and "High Confidence" (40.15%).
- **Sub-Cluster 0 (Coastal Debris or mixed with Sediment):** Presents tendentially neutral spectral and texture values (slightly negative FDI and NDVI), but is the only group to record a positive peak in the BSI (Bare Soil Index, 0.65). This suggests beached plastic, in very shallow waters or mixed with strong sedimentary turbidity, states for which annotators still feel confident (*High Confidence* of 52.14%).

The 2D visualization of the sub-clusters via a two principal components PCA is depicted in Figure 10, while the values of the centroids are shown in Figure 9.

4.2 Classification

For the classification phase, several columns were removed from the dataset: `coordinates`, `image`, `tile`, and `date`. Subsequently, the dataset was divided into training and testing sets using an 80/20 split. The

Sub-Cluster	Annotator Confidence	Pixels Count	Confidence (%)
Sub-Cluster 0 (Coastal/Sediment - 30%)	High	499	52.14
	Moderate	347	36.26
	Low	111	11.60
Sub-Cluster 1 (Dense/Pure - 18%)	High	340	64.89
	Moderate	147	28.05
	Low	37	7.06
Sub-Cluster 2 (Sparse/Submerged - 29%)	Moderate	343	40.35
	High	257	30.24
	Low	250	29.41
Sub-Cluster 3 (Mixed w/ Veg. - 23%)	Moderate	288	43.97
	High	263	40.15
	Low	104	15.88

Table 4: Distribution of annotator confidence within the Marine Debris sub-clusters ($k = 4$).

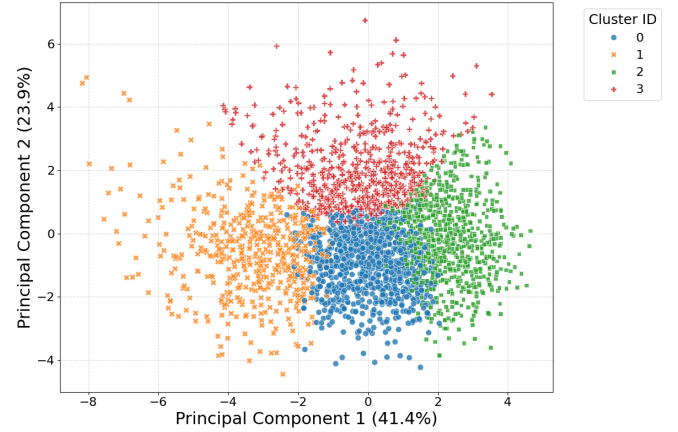


Figure 10: 2D visualization of the clustering results for the clustering on Marine Debris pixels, projected onto the first two Principal Components (PCA). The percentages in parentheses represent the explained variance.

Confidence column was indexed, and its **high**, **moderate**, and **low** values were transformed using one-hot encoding. Following this preprocessing, three distinct models were trained and evaluated.

The first model is an XGBoost classifier. Hyperparameter tuning was performed using a random search with 8 combinations and 5-fold cross-validation across the following parameter space: `max_depth` [4, 6, 8], `subsample` [0.7, 0.8, 1.0], and `n_estimators` [50, 100].

Using the best resulting XGBoost model (`max_depth` = 6, `n_estimators` = 50, `subsample` = 0.8) on the test set yielded to the following results (Table 5):

As expected, heavily represented classes such as **Sed-Lad Water** and **Turbid Water** achieve very high F1-scores. When classes are sorted by support in descending order, a clear decreasing trend in F1-score emerges, confirming the positive correlation between classifier performance and class support. Among the low-support classes, however, some notable exceptions stand out: **Dense Sargassum** and **Foam** both achieve remarkably high F1-scores (in

Class	Support	Precision	Recall	F1-Score
Sed-Lad Water	74532	> 0.99	1.00	1.00
Turbid Water	31392	0.99	0.99	0.99
Marine Water	24865	0.95	0.97	0.96
Clouds	18863	0.97	0.98	0.97
Shallow Water	3240	0.96	0.94	0.95
Cloud Shadows	2275	0.88	0.75	0.81
Wakes	1581	0.80	0.63	0.71
Waves	1065	0.83	0.79	0.81
Marine Debris	609	0.87	0.89	0.88
Ship	298	0.73	0.71	0.72
Sparse Sargassum	276	0.84	0.81	0.82
Nat Org Material	142	0.85	0.73	0.79
Mixed Water	77	0.37	0.13	0.19
Dense Sargassum	33	0.88	0.88	0.88
Foam	21	0.93	0.67	0.78

Table 5: Per-class classification metrics of the XGBoost multiclass baseline model, sorted by support in descending order.

order: 0.88 and 0.78) despite their very limited representation, suggesting that these classes exhibit highly distinctive spectral patterns that the model can reliably detect. In contrast, classes such as **Mixed Water** lack this strong discriminative signature, resulting in comparatively lower performance.

Regarding our target class, **Marine Debris**, the model achieves a relatively strong F1-score of 0.88, which is particularly encouraging given the small number of support samples available for this class.

To further investigate the classification of **Marine Debris** and maximize the classifier’s performance on this class of interest, the target variable was subsequently binarized, assigning a label of 1 to all **Marine Debris** records and 0 to all remaining classes.

Using the same hyperparameter tuning grid explored by random search with 8 combinations and 5-fold cross-validation, we achieved the second model of our study, the results of which are reported in Table 6.

Class	Support	Precision	Recall	F1-Score
0 (Other)	158660	> 0.99	> 0.99	> 0.99
1 (M. Debris)	609	0.90	0.90	0.90

Table 6: Classification metrics of the XGBoost binary classifier (**Marine Debris** vs. all other classes).

The third model evaluated was a Random Forest classifier. A similar tuning approach was applied using a random search (8 combinations, 5-fold cross-validation) across the following parameters: `max_depth` [4, 6, 8], `subsample_rate` [0.7, 0.8, 1.0], and `n_trees` [50, 100]. The optimal parameters found based on the F1-score were a `max_depth` of 8, a `subsample_rate` of 0.8,

and 100 trees.

However, this model underperformed compared to XGBoost, achieving a general F1-score of 0.92 and a specific F1-score of 0.79 on the **Marine Debris** class.

4.2.1 Imbalanced Learning

The class imbalance that emerged during the initial EDA, and subsequently confirmed in Section 4.2, motivated the exploration of several imbalanced learning techniques.

The dominant class (**Sediment-Laden Water**, 74,532 samples) exceeds the target class **Marine Debris** (609 samples) by a ratio of approximately 122:1, and **Foam** with only 21 samples by nearly 3,500:1. This distribution makes it trivial to achieve high global accuracy but difficult to correctly classify the rare classes.

Before examining the results, an important infrastructural constraint must be highlighted: unlike environments such as Pandas or Scala, PySpark does not provide native libraries or frameworks for handling class imbalance. For this reason, well-established but unofficial implementations of two algorithms were adopted, namely **ADASYN**² and **Cluster Centroids**³, and adapted to the distributed computing context.

From an implementation standpoint, both algorithms proved to be the most adaptable to the PySpark ecosystem. ADASYN was implemented by computing k -nearest neighbors via `sklearn` on a stratified sample of the dataset, a necessary workaround in the absence of a native distributed KNN library, and then generating synthetic samples that were reinjected into the Spark DataFrame. Cluster Centroids, conversely, could directly leverage `pyspark.ml.clustering.KMeans`, a native MLlib component, making the undersampling process fully distributed and scalable without any external dependencies.

Due to the lack of a native distributed k -Nearest Neighbors implementation in PySpark MLlib, the KNN computation in ADASYN was offloaded to

`sklearn.NearestNeighbors`, applied on a stratified sample of the dataset (`sample_fraction=0.1`, with a minimum of 10,000 points). This design choice avoids cache and memory overflow at distributed scale while preserving fidelity to the original algorithm. The resulting hybrid strategy ensures scalability on large datasets without sacrificing ADASYN’s core adaptive property.

The Imbalanced Learning techniques were applied to both the multiclass and binary classification settings. In the binary case (1 = **Marine Debris**, 0 = **Others**), the objec-

²<https://github.com/fsleeman/spark-class-balancing>

³<https://github.com/ElsevierSoftwareX/SOFTX.2019.253>

tive was to maximize performance on the class of interest by fully exploiting the potential of rebalancing techniques. In the multiclass case, the informational value of error patterns was preserved, specifically, which classes are most commonly confused with **Marine Debris**, yielding more modest performance gains, but providing richer diagnostic insights for the analysis.

All models discussed in this subsection were trained using XGBoost as the base classifier, following the same preprocessing pipeline, train/test split, and random search over the same hyperparameter space described in Section 4.2. For both ADASYN and ClusterCentroids, a target threshold of 10,000 samples was used. In the oversampling setting, all classes below this threshold were upsampled to 10,000, while classes already above it were left unchanged. In the undersampling setting, all classes exceeding this threshold were downsampled to 10,000, while smaller classes were left unchanged.

Model	Precision	Recall	F1-Score
XGBoost Multi	0.98	0.98	0.98
RandomForest Multi	0.94	0.94	0.92
ADASYN Multi	0.99	0.99	0.99
ClusterC. Multi	0.94	0.89	0.90
XGBoost Binary	> 0.99	> 0.99	> 0.99
ADASYN Binary	> 0.99	> 0.99	> 0.99
ClusterC. Binary	> 0.99	> 0.99	> 0.99

Table 7: Weighted Precision, Recall, and F1-Score across all trained models for both the multiclass and binary settings. All models except Random Forest use XGBoost as the base classifier. Bold entries indicate the best-performing model within each setting.

In Table 7, among multiclass classifiers, XGBoost with ADASYN oversampling achieves the highest scores across all three metrics (Precision: 0.99, Recall: 0.99, F1-Score: 0.99), outperforming both the baseline XGBoost Multi and the ClusterCentroids undersampling variant. The latter, despite maintaining a competitive weighted F1-Score of 0.91, exhibits a significant drop in recall, reflecting its tendency to lose discriminative structure for medium-frequency classes when aggressively downsampling the dominant ones.

In the binary classification scenario, all three models achieve near-perfect weighted metrics (F1-Score > 0.99), a result that is largely driven by the extreme class imbalance: with over 99.6% of the test set belonging to the non-debris class, high weighted scores are trivially attainable without effectively detecting **Marine Debris**.

Table 8 instead presents a comparison of all trained mod-

Model	Precision	Recall	F1-Score
XGBoost Multi	0.87	0.89	0.88
RandomForest Multi	0.80	0.78	0.79
ADASYN Multi	0.94	0.89	0.91
ClusterC. Multi	0.41	0.73	0.53
XGBoost Binary	0.90	0.90	0.90
ADASYN Binary	0.87	0.88	0.87
ClusterC. Binary	0.79	0.99	0.88

Table 8: Precision, Recall, and F1-Score computed exclusively on the **Marine Debris** class across all trained models. All models except Random Forest use XGBoost as the base classifier. Bold values indicate the best Precision, best Recall, and best F1-Score respectively.

els using metrics computed exclusively on the **Marine Debris** class.

Multiclass XGBoost with ADASYN achieves the best F1-score (0.91): precision jumps from 0.87 to 0.94 while Recall remains unchanged at 0.89, suggesting that the synthetic samples generated by ADASYN refined the decision boundary without introducing additional noise. Binary XGBoost with ClusterCentroids, on the other hand, achieves a remarkably high recall (0.99), which is particularly relevant when the application objective is to maximize detection rate, even at the cost of a higher number of false positives.

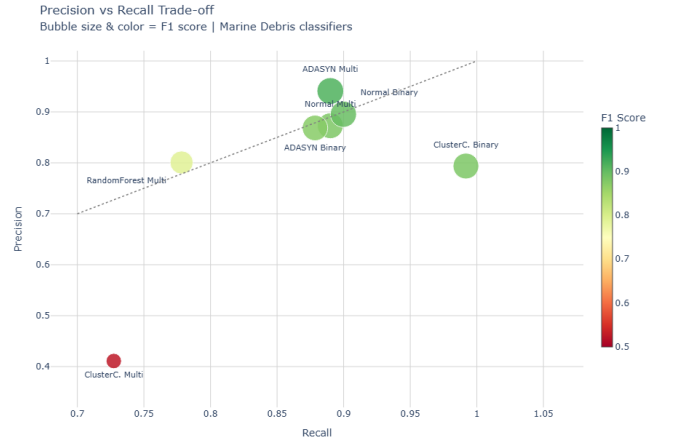


Figure 11: Scatter plot of Precision vs. Recall for **Marine Debris** detection across all seven model configurations (multiclass and binary). Each bubble represents one model; size and color encode the corresponding F1-Score on the **Marine Debris** class, ranging from low (red) to high (dark green). The dashed line represents the iso-F1 diagonal.

Figure 11 illustrates the precision-recall trade-off for **Marine Debris** detection across all configurations. Multiclass XGBoost with ADASYN achieves the best balance, with the highest F1-Score (0.91) driven by a substantial gain in precision (+0.068 over the baseline) while maintaining recall at 0.89. Binary XGBoost with ClusterCentroids

occupies the opposite end of the spectrum: it maximises recall (0.99) at the cost of a significantly lower precision (0.79), making it suitable in contexts where minimising false negatives is the primary objective. The baseline multiclass and binary models cluster closely together, confirming that simple binarisation alone provides marginal benefit over the standard multiclass approach.

4.2.2 Domain Shift Analysis

This analysis evaluates the model’s ability to generalize spatially, ensuring that it learns universal spectral signatures rather than local background features. The dataset was split using a strict geographic criterion: the source domain (training set) comprises the pixels from Central America (738,758 samples), while the target domain (test set) includes the rest of the world (57,197 samples). To prevent spatial data leakage, coordinates and Tile IDs were removed.

To handle the strong class imbalance, in this case, we tested variants of the model with and without class weights. In the weighted models, balancing was performed by assigning each class a weight inversely proportional to its frequency (support) in the training set, like in the MARIDA benchmark paper.

XGBoost hyperparameter optimization was performed solely on the training set via Random Search, with a 3-fold cross-validation approach, exploring the following ranges: `max_depth` $\in \{4, 6, 8\}$, `learning_rate` $\in \{0.05, 0.1, 0.2\}$, `subsample` $\in \{0.7, 0.8, 1.0\}$ and `n_estimators` $\in \{50, 100\}$. To address the extreme class imbalance, the optimization was guided by the F1-Score.

In the 15-class task, the weighted model showed a severe vulnerability to domain shift. For instance, to minimize the weighted error, it misclassified 1219 pixels of **Sediment-Laden Water** (SLW) as **Marine Debris** (MD). This happens because sediments are the most abundant class in the training set (low weight): the model therefore “sacrificed” the prediction of mud in favor of plastic, causing the Precision of MD to plummet to 0.21.

By removing the weights, the algorithm was forced to accurately separate the two materials: false positives from SLW dropped from 1219 to 8, bringing the Recall of SLW to 1.00 and the F1-Score of MD to 0.65. However, this approach nullified the ability to detect rare classes such as **Foam** and **Mixed Water**. Table 9 summarizes this trade-off.

Following the same approach adopted in the other classification sections, the problem was then reformulated as a

Model	Marine Debris			Sediment-Laden Water		
	Prec	Rec	F1	Prec	Rec	F1
Weighted	0.21	0.88	0.33	0.96	0.23	0.38
Unweighted	0.55	0.79	0.65	0.98	1.00	0.99

Table 9: Multi-class classification metrics for the domain shift analysis. Performance comparison between Marine Debris and Sediments to highlight the impact of weights.

binary task (**Marine Debris** vs. **Other**).

As shown in Table 10, the binary weighted model achieved an exceptional Recall (0.90), ideal for highly sensitive ecological alert systems that must not miss any pollution. In contrast, the binary unweighted model provided the most robust compromise, maximizing Precision (0.66) and F1-Score (0.71), proving to be the optimal choice for reliable and automated global monitoring, with few false alarms.

Model	Prec	Rec	F1
Weighted	0.54	0.90	0.68
Unweighted	0.66	0.77	0.71

Table 10: Binary classification metrics for the domain shift analysis.

4.2.3 Final Results

The experimental results demonstrate that XGBoost with ADASYN oversampling consistently outperforms all baseline models proposed in the MARIDA benchmark paper (Kikaki et al., 2022) across all relevant metrics. For Marine Debris detection specifically, our best multiclass classifier achieves an F1-Score of 0.91, compared to the best result reported by Kikaki et al. of 0.80 (RF with spectral signatures and spectral indices), representing an improvement of approximately 11 percentage points. At the macro-averaged level, our ADASYN Multi configuration achieves a macro F1-Score of 0.85 over 15 classes, against the 0.79 reported by their best overall model (RF with spectral signatures, indices and GLCM features) evaluated on an aggregated 11-class version of the same dataset. It is worth noting that this comparison should be interpreted with caution: MARIDA’s baselines reduce the original 15 classes to 11 by merging spectrally similar water-related categories, which simplifies the classification task and tends to inflate macro-averaged scores. Our models, trained and evaluated on the full 15-class schema, face a strictly harder problem while still achieving superior results. Furthermore, the binary classification approach, while not directly comparable to MARIDA’s multiclass setup, achieves near-perfect weighted metrics (F1 > 0.99),

confirming the feasibility of a dedicated **Marine Debris** detector when the task is framed as a one-vs-rest problem. Overall, the adoption of gradient boosting with targeted oversampling via ADASYN proves to be a robust and effective strategy for **Marine Debris** detection in multi-spectral remote sensing data, surpassing both shallow and deep learning baselines established on the same benchmark dataset.

Comparing the outcomes of Domain Shift analysis with the standard classification task reveals critical insights. The excellent metrics achieved in the random 80/20 split are over-optimistic due to the intrinsic nature of satellite imagery. This is caused by spatial autocorrelation: nearby pixels share identical atmospheric conditions, solar illumination, and water turbidity. Consequently, a random split inherently introduces spatial data leakage. The model memorizes local environmental backgrounds instead of learning the universal spectral signature of marine debris. Therefore, the performance drop observed during the Domain Shift analysis is not a model failure. Instead, it reveals the model’s true generalization power. Changing regions means facing unseen seabed compositions and new water types. Evaluating the model on completely unseen geographic areas is the only reliable way to assess its actual operational readiness on a global scale.

4.3 Explainable AI

In this phase, we conducted an Explainable AI (XAI) analysis on the best classification model (XGBoost oversampled with ADASYN), extracting both the global feature importance and the SHAP summary plot. The reference framework for distributed XAI in PySpark is **SynapseML**⁴; however, it requires every pipeline stage to implement the `_to_java()` interface for JVM serialisation, which `SparkXGBClassifierModel` does not support. SHAP values were therefore computed via `shap.TreeExplainer` on the underlying booster (`get_booster()`), distributed across workers using a PySpark `pandas_udf`, and stored as columns of a Spark DataFrame throughout.

The analysis of the Summary Plot and the mean SHAP importance reveals how the XGBoost model makes its decisions. Below are the main observations regarding the most influential features:

- **FDI (Floating Debris Index):** With a mean importance of 1.71, this index, specifically designed to detect floating material, is the most discriminating

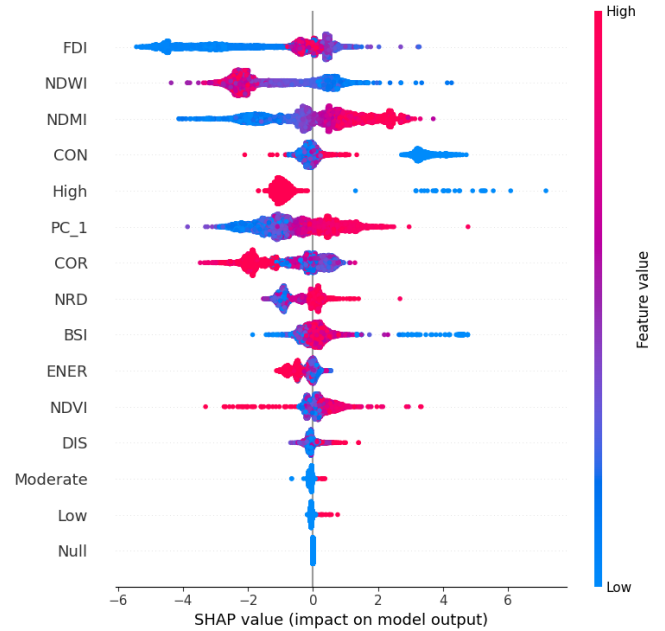


Figure 12: Summary Plot of SHAP values for **Marine Debris** class (sample of 8000 records).

parameter for the model. The swarm plot shows a clear pattern: high values of FDI (represented by red dots) have a moderate to strong negative impact on the SHAP value, while low values (blue dots) spread widely on the left side. The model heavily relies on this index to separate the target class from the background.

- **NDWI (Normalized Difference Water Index):** Ranking second with a mean importance of 1.50, this optical parameter remains crucial. The plot shows that high values of this index (red dots) have a strong negative impact (pushing the prediction away from the target class), while lower values (blue dots) push the prediction towards the right. This makes physical sense: pure water has high NDWI, while the presence of plastic, nets, or floating debris alters this spectral signature, causing a drop in NDWI values that the model recognizes as an anomaly.
- **NDMI (Normalized Difference Moisture Index) & CON (Contrast):** Ranking third (1.14) and fourth (1.13) respectively, these features show significant influence. For NDMI, high values (red dots) strongly push the prediction to the right (positive SHAP value), indicating that the moisture profile is a key indicator. Interestingly, CON (a texture metric) shows that lower values (blue dots) are strongly associated with positive SHAP values, suggesting that a specific textural smoothness or uniformity is linked to the model’s detection of de-

⁴<https://github.com/microsoft/SynapseML>

bris.

- **The "High" variable (Confidence):** Ranking fifth (0.99), this category derived from One-Hot Encoding shows an interesting dynamic. The plot displays a tight cluster of high values (red dots) with a negative SHAP impact, but a long tail of low values (blue dots, meaning the original "High" label was absent) associated with positive SHAP values. This suggests that the model often classifies pixels as **Marine Debris** based on their optical and textural data, even when the initial human-provided confidence labels were not "High".
- **PC_1 (Principal Component 1):** In sixth place (0.98), this synthetic variable derived from the Principal Component Analysis maintains its relevance. Both high (red) and low (blue) values spread across both sides of the zero line, indicating a complex, non-linear relationship where the optical combination captured by this component acts as a robust, albeit multifaceted, marker for identifying debris.
- **NDVI (Normalized Difference Vegetation Index):** Even though it ranks lower in overall mean importance (0.31), the vegetation index shows a very relevant non-linear pattern. The plot highlights a long right tail made of red dots: when NDVI values are exceptionally high, the model's certainty increases drastically. This behavior can be linked to the physical reality of ocean debris, which often gets colonized by microalgae or tangles with marine vegetation, creating a strong "green" spectral signature.

5 Conclusions

This work demonstrated that distributed analysis of multispectral satellite imagery via Apache Spark is not only technically feasible, but practically effective for the detection and profiling of marine debris at scale. By combining spectral indices, GLCM texture features, and a PCA-based dimensionality reduction pipeline, we constructed a feature space rich enough to support both unsupervised exploration and high-performance supervised classification on the full 15-class MARIDA dataset.

The clustering analysis revealed that the global spectral structure of the data does not naturally align with the semantic taxonomy defined by the annotators: even at the optimal configuration of $k = 6$, **Marine Debris** cannot form a pure cluster, with its pixels inevitably ab-

sorbed by floating vegetation and surface reflectance confounders. The targeted sub-clustering on **Marine Debris** pixels alone, however, yielded a physically meaningful taxonomy of four macro-states, each validated by the annotator confidence distribution and directly interpretable in terms of spectral and textural signatures.

On the classification side, our best model, XGBoost with ADASYN oversampling, achieves an F1-Score of 0.91 on **Marine Debris** in the multiclass setting, representing an improvement of approximately 11 percentage points over the best MARIDA baseline (Random Forest, $F1 = 0.80$). This result is particularly significant given that our models operate on the full 15-class schema, while the MARIDA baselines were evaluated on a simplified 11-class version, making our task strictly harder. The XAI analysis further confirmed the physical interpretability of the model, with FDI, NDWI, and NDMI emerging as the most discriminative features, consistent with the physical properties of floating debris established during the EDA.

The Domain Shift Analysis exposed the limits of standard random splits in geographically concentrated datasets: the performance drop observed when training on Central America and testing on the rest of the world is not a model failure, but a precise quantification of spatial overfitting induced by the dataset's intrinsic geographic bias. The binary unweighted model, achieving an F1-Score of 0.71 on completely unseen regions, represents the most operationally honest estimate of global generalization capability.

Future work should address three main directions. First, Domain Adaptation techniques, such as instance reweighting or adversarial feature alignment, should be integrated into the PySpark pipeline to reduce the performance gap between seen and unseen geographic regions, moving towards a globally robust model capable of operating across diverse oceanic environments. Second, the integration of Synthetic Aperture Radar (SAR) data from Sentinel-1 could overcome the critical limitation imposed by cloud cover, which currently renders optical-only models blind to a significant fraction of acquisition conditions; the fusion of SAR and multispectral features within a distributed pipeline represents a natural and high-impact extension of this work. Third, the four physical sub-states identified by the sub-clustering analysis: dense, sparse, vegetation-mixed, and sediment-mixed debris, suggest the feasibility of a fine-grained hierarchical classifier that, rather than treating **Marine Debris** as a monolithic class, distinguishes between its physical manifestations, enabling more targeted and actionable monitoring outputs.